

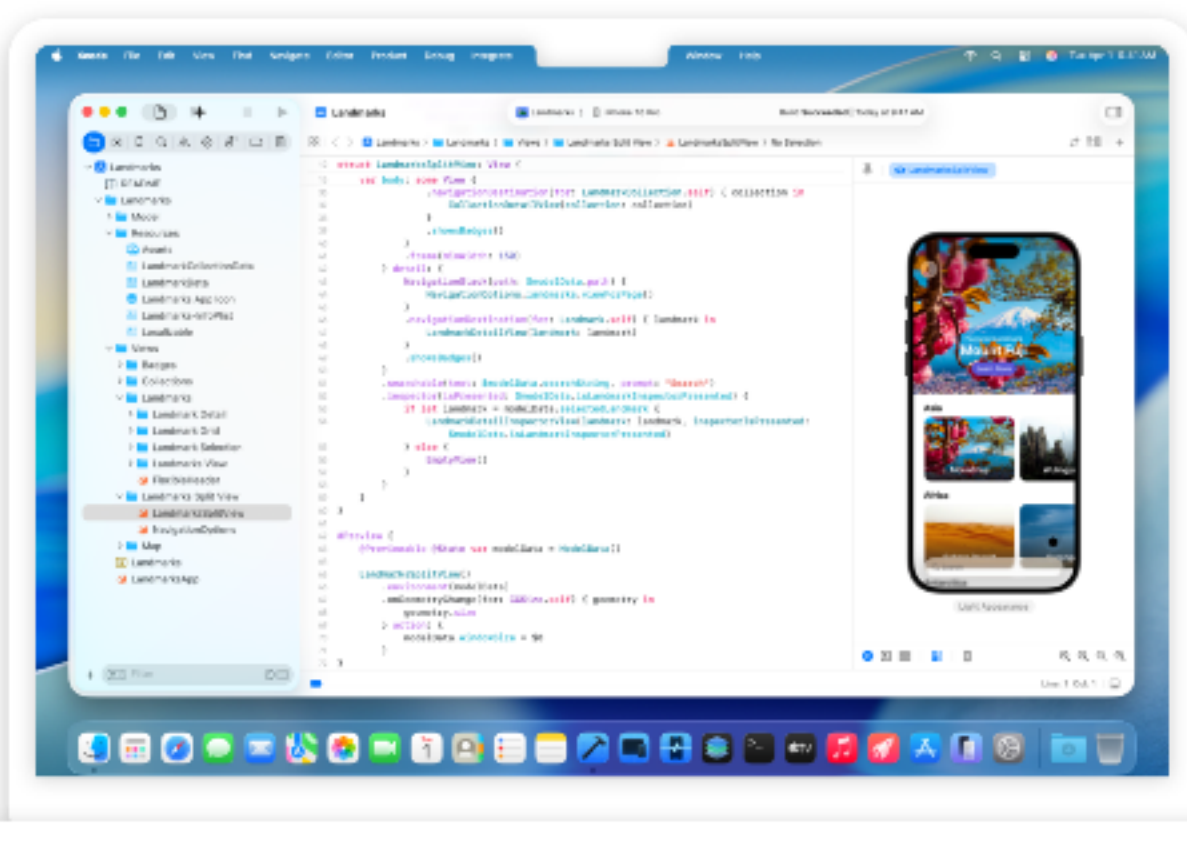
Documentation

Xcode

Build, test, and submit your app with Apple’s integrated development environment.

Overview

Xcode is the suite of tools you use to build apps for Apple platforms. Use Xcode to manage your entire development workflow — from creating your app to testing, optimizing, and submitting it to the App Store.



Xcode includes a world-class source editor with code completion, source control, and a

powerful debugger. Use coding intelligence to explain and write code, analyze bugs, and generate fixes. Add playground macros to run code snippets, and add SwiftUI previews to see your UI as you build it.

Xcode also includes several tools to help you rapidly develop and test your app:

- Run your entire app in Simulator without using a real device.
- Create a single, multilayer icon for your app using the Icon Composer app.
- Use Instruments to profile and analyze your app, improve performance, and investigate system resource usage.
- Construct 3D content with Reality Composer.
- Train custom machine learning models with Create ML.
- Identify areas of your app that aren't accessible with Accessibility Inspector.

Note

Download the latest version of Xcode from the [Mac App Store](#). Download beta versions of Xcode from the [Apple Developer website](#).

Topics

Essentials



Creating an Xcode project for an app

Start developing your app by creating an Xcode project from a template.



Creating your app's interface with SwiftUI

Develop apps in SwiftUI with an interactive preview that keeps the code and layout in sync.



Previewing your app's interface in Xcode

Iterate designs quickly and preview your apps' displays across different Apple devices.



Building and running an app

Compile your source files and assemble an app bundle to run on a device or simulator.



Xcode updates

Learn about important changes to Xcode.

Xcode IDE



Projects and workspaces

Manage the code and resources you use to build apps, libraries, and other software for Apple platforms.



Source control management

Back up your files, collaborate with others, and tag your releases with Git source control support in Xcode.



Capabilities

Enable services that Apple provides, such as In-App Purchase, Push Notifications, Apple Pay, iCloud, and many others.



Build system

Compile your code into a binary format, and customize your project settings to build your code.



Command-line tools

Develop and customize your projects in Terminal.

Code



Source editor

Edit your source files, locate issues, and make necessary changes using the source editor.



Bundles and frameworks

Organize code and resources in bundles and frameworks.

☰ Swift packages

Create reusable code, organize it in a lightweight way, and share it across Xcode projects and with other developers.

Interface

☰ Asset management

Add app icons, images, strings, data files, machine learning models, and other resources to your projects, and manage how you load them at runtime.

☰ Localization

Expand the market for your app by supporting multiple languages and regions.

☰ Accessibility Inspector

Reveal how your app represents itself to people using accessibility features.

Documentation

☰ Writing documentation

Produce rich and engaging developer documentation for your apps, frameworks, and packages.

Tuning and debugging

☰ Devices and Simulator

Configure and manage devices connected to your Mac or devices in Simulator and use them to run your app.

☰ Debugging

Identify and address issues in your app using the Xcode debugger, Xcode Organizer, Metal debugger, and Instruments.

☰ Performance and metrics

Measure, investigate, and address the use of system resources and issues impacting performance using Instruments and Xcode Organizer.

 Testing

Develop and run tests to detect logic failures, UI problems, and performance regressions.

Distribution and continuous integration

 Distribution

Prepare your app and share it with your team, beta testers, and customers.

 Xcode Cloud

Automatically build, test, and distribute your apps with Xcode Cloud to verify changes and create high-quality apps.

Hardware considerations

 Apple silicon

Get the resources you need to create software for Macs with Apple silicon.

 Application binary interfaces

Write assembly instructions that adhere to the application binary interfaces of Apple platforms.